A Uniform Geometric-Embedding Threshold for Trees of Moderately Growing Degree

FA/Banach run `fa_banach_001`

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Status

This packet records a candidate substantial partial answer to Problem 1.6 of Altschuler–Dodos–Tikhomirov–Tyros [1]. The proof is complete modulo the estimates already proved in that source, and the current verdict is

$$\boxed{\textbf{likely valid partial result; human review requested.}}$$

It does not settle the full growing-degree problem or the optimal-constants Problem 1.5.

Source problem

A graph $G = (V, E)$ is *geometrically embeddable* into a normed space X if there is a map $\zeta : V \to X$ such that, for all distinct $u, v \in V$,

$$\|\zeta(u) - \zeta(v)\|_X \leq 1 \quad \iff \quad \{u, v\} \in E.$$

The source proves a universal $\Theta(\log N/\log\log N)$ threshold when the maximum degree Δ is fixed. On PDF page 3 it asks for the threshold when Δ grows with N.

Problem 1.6 (Trees of growing maximal degree)**.** *Determine an explicit function $m := m(N, \Delta)$ and some universal constants α_0 and α_1 so that the following holds. Any tree on N vertices with maximum degree Δ embeds into any normed space X of dimension at least $\alpha_1 m$. And, no complete tree of degree Δ embeds into any space of dimension less than $\alpha_0 m$.*

Proof overview. Part (A) of Theorem 1.1 is essentially due to Frankl and Maehara. The argument is volumetric: in low dimension there is insufficient volume to pack the required number of disjoint translates of a fixed fraction of the unit ball. An explicit witness is given by the complete rooted $(\Delta - 1)$–regular tree of appropriate height.

We now outline the proof of part (B), which is our main contribution. Let $X = (\mathbb{R}^m, \|\cdot\|_X)$

In transcription, Problem 1.6 asks for an explicit function $m(N, \Delta)$ and universal constants α_0, α_1 such that every N-vertex tree of maximum degree Δ embeds into every normed space of dimension at least $\alpha_1 m$, while a complete tree of degree Δ does not embed below $\alpha_0 m$.

Proof Intuition

The height of a complete degree-Δ tree with N vertices is naturally

$$H = \frac{\log N}{\log(\Delta - 1)}.$$

Packing its mutually nonadjacent vertices forces dimension of order $\log N/\log(2+H)$. This suggests the scale

$$q(N, \Delta) = \frac{\log N}{\log(2 + H)}.$$

The upper bound is hidden in the source's randomized construction. Its local lemma separates paths into three ranges. Short paths are controlled with probability $\exp(-m^{\kappa/2})$ and have dependency cost $\exp(O(\log \Delta))$; this is the origin of the growth assumption below. For paths of length at most H, the largest logarithmic dependency cost is

$$\frac{H}{\log H} \log \Delta \asymp \frac{\log N}{\log H} \asymp q.$$

For paths longer than H, the probability exponent is $m \log H \asymp \log N$. Thus $m = 64q$ closes every local-lemma inequality. The Gaussian part of the source proof contains no Δ and remains uniform on the whole interval $c \log N/\log\log N \leq m \leq C \log N$.

The partial theorem

Let $\kappa \in (0, 1)$ be the absolute exponent in the thin-shell estimate stated as Theorem 2.2 of [1].

Theorem 1 (Uniform moderately growing-degree threshold)**.** *Let (N_j) and (Δ_j) be integer sequences such that $N_j \to \infty$ and $3 \leq \Delta_j \leq N_j - 1$. Set*

$$D_j = \log(\Delta_j - 1), \qquad H_j = \frac{\log N_j}{D_j}, \qquad q_j = \frac{\log N_j}{\log(2 + H_j)}.$$

Assume

$$D_j = o\big(q_j^{\kappa/2}\big). \tag{1}$$

Then, for all sufficiently large j, the following hold.

(i) Every tree on N_j vertices of maximum degree at most Δ_j geometrically embeds into every normed space of dimension at least $\lceil 64q_j \rceil$.

(ii) There is a breadth-first truncation of the complete rooted degree-Δ_j tree on N_j vertices which does not geometrically embed into any normed space of dimension at most $q_j/2$.

At cardinalities of a full complete rooted tree, the witness in (ii) is the full complete tree. Hence $q(N, \Delta)$ is the threshold scale, up to universal constants, throughout the regime (1)*.*

Corollary 2. *For every fixed $\beta < \kappa/2$, Theorem 1 applies whenever*

$$3 \leq \Delta_N \leq \exp\big((\log N)^\beta\big).$$

Proof

We suppress the sequence index. Put $D = \log(\Delta - 1)$, $H = \log N/D$, and $q = \log N/\log(2 + H)$.

Elementary consequences of the growth condition

Since $D \geq \log 2$, we have

$$H \leq \frac{\log N}{\log 2}, \qquad q \geq c\frac{\log N}{\log\log N}, \qquad q \leq C \log N$$

for absolute constants $c, C > 0$. Conversely $q \leq C \log N$, so (1) implies $D = o((\log N)^{\kappa/2})$. Therefore $H = \log N/D \to \infty$, and

$$\frac{\log H}{\log(2+H)} \longrightarrow 1. \tag{2}$$

Retain the source parameters

$$c_1 = \frac{1}{4}, \qquad c_2 = \min\left\{\frac{1}{4}, \frac{\kappa}{4}\right\}, \qquad \ell_0 = \exp\left((\log N)^{c_2}\right),$$

and its absolute integer k_0. Set

$$m = \lceil 64q \rceil, \qquad k_1 = \lfloor H \rfloor.$$

Because $H \to \infty$ but $H = O(\log N)$, while ℓ_0 grows faster than every power of $\log N$, eventually

$$k_0 < k_1 < \ell_0. \tag{3}$$

Also

$$(\log N)^{c_2} = o(q^{\kappa/2}) \quad \text{and} \quad (\log N)^{c_2} = o(q). \tag{4}$$

Indeed, $c_2 \leq \kappa/4$ and $q \geq c \log N/\log\log N$.

Uniform local-lemma estimate

Use exactly the random edge vectors, path events, dependency graph, and weights from Section 4.2 of [1]. For a path-event of length k, the weight is

$$b_k = \begin{cases} \dfrac{1}{2^{k+1}6\ell_0 k(\Delta-1)^k}, & 2 \leq k \leq k_1, \\ \dfrac{3}{2\pi^2 k^2 N^2}, & k_1 < k \leq \ell_0. \end{cases}$$

Claim 4.1 of the source bounds the number of dependent length-k paths for a fixed path by the minimum of N^2 and a constant multiple of $\ell_0 k(\Delta-1)^k$. If $k \leq k_1$, then $(\Delta-1)^k \leq N$, while $k\ell_0 = N^{o(1)}$. Hence, uniformly,

$$6k\ell_0(\Delta-1)^k \leq N^2 \tag{5}$$

for all sufficiently large N. The product estimate (4.16) of the source therefore remains valid:

$$\prod_{B\sim A}(1-b_B) \geq \frac{1}{2} \tag{6}$$

for every path-event A. It remains to compare each event probability with $b_k/2$.

For $2 \leq k \leq k_0$, Lemma 3.3 of the source gives

$$\mathbb{P}(A) \leq \exp(-m^{\kappa/2}).$$

On the other hand,

$$\log \frac{2}{b_k} leq C_0 + k_0 \log(2\Delta - 2) + (\log N)^{c_2} = o(q^{\kappa/2}) = o(m^{\kappa/2})$$

by (1) and (4). Thus $\mathbb{P}(A) \leq b_k/2$ eventually.

Next suppose $k_0 < k \leq k_1$. Lemma 3.2 of the source, with $c_1 = 1/4$, gives

$$\mathbb{P}(A) \leq \exp\left(-\frac{m}{8} \log k\right). \tag{7}$$

The function $x/\log x$ is increasing for $x \geq 3$. Since $\log(2\Delta - 2) = D + \log 2 \leq 2D$, (2) gives

$$\begin{aligned} \frac{k}{\log k} \log(2\Delta - 2) &\leq \frac{k_1}{\log k_1} \log(2\Delta - 2) \\ &\leq (2 + o(1)) \frac{HD}{\log H} = (2 + o(1)) \frac{\log N}{\log H} = (2 + o(1))q. \end{aligned} \tag{8}$$

The constant, $\log k$, and $\log \ell_0$ terms in $\log(2/b_k)$ contribute $o(q) \log k$ by (4). Consequently

$$\log \frac{2}{b_k} \leq (3 + o(1))q \log k,$$

whereas $m/8 \geq 8q$. Comparing with (7) yields $\mathbb{P}(A) \leq b_k/2$ uniformly in this range.

Finally suppose $k_1 < k \leq \ell_0$. The same small-ball estimate gives

$$\mathbb{P}(A) \leq \exp\left(-\frac{m}{8} \log k\right) \leq \exp\left(-8q \log H\right) = \exp\big(-(8 - o(1)) \log N\big),$$

using (2). Meanwhile

$$\log \frac{2}{b_k} = 2 \log N + 2 \log k + O(1) = (2 + o(1)) \log N$$

because $k \leq \ell_0 = N^{o(1)}$. Again $\mathbb{P}(A) \leq b_k/2$.

Together with (6), these three estimates verify the asymmetric Lovasz local lemma [2]. Thus the source's event $\mathcal{L}$, on which every nonadjacent pair at tree distance at most ℓ_0 has the required uniform-vector separation, has positive probability.

Uniform Gaussian completion

The remainder of Section 4.3 of [1] is independent of Δ. We record why every "sufficiently large" estimate there is uniform for the new dimension. From the elementary bounds above,

$$c \frac{\log N}{\log \log N} \leq m \leq C \log N. \tag{9}$$

The source uses

$$\delta = \exp\left(-\left(\frac{1}{2} - \frac{c_1}{2}\right) (\log N)^{c_2}\right).$$

For adjacent pairs its Gaussian tail threshold is $m^{-1/2}\delta^{-1}(\log N)^{-1}$. For distances $2 \leq k \leq k_0$ it is, up to absolute factors, $\delta^{-1}m^{-(3/2-\kappa)}$. For $k_0 < k \leq \ell_0$ it is, up to absolute factors, $\exp((c_1/2)(\log N)^{c_2})m^{-1/2}$. In all three cases the exponential in $(\log N)^{c_2}$ dominates every power of

m and $\log N$ allowed by (9). The Gaussian upper-tail estimate therefore makes each bad probability at most N^{-2}, exactly as in the source.

For $k > \ell_0$, the source's Gaussian small-ball bound is

$$\exp\big(-\tfrac{c_1}{4}m(\log N)^{c_2}\big).$$

The lower bound in (9) implies $m(\log N)^{c_2} \gg \log N$, so this is also at most N^{-2}.

Conditioned on $\mathcal{L}$, fewer than $N^2/2$ unordered pairs remain. The union bound is therefore strictly less than $1/2$, and a geometric embedding exists with positive probability. If the ambient normed space has dimension larger than m, perform the construction in any m-dimensional subspace. This proves part (i).

Complete-tree obstruction

Use the breadth-first truncated complete tree from part (A) of the source's Theorem 1.1. It has N vertices, maximum degree at most Δ, a root r, and radius

$$h_0 = \left\lceil \frac{\log\left((N-1)(\Delta-2)/\Delta+1\right)}{\log(\Delta-1)} \right\rceil \le H+1.$$

At full complete-tree cardinalities, no truncation occurs.

Suppose it geometrically embeds into a d-dimensional normed space X. Because the tree is bipartite, it has an independent set V_0 with $|V_0| \ge N/2$. The radius-$1/2$ balls centered at the images of vertices in V_0 are pairwise disjoint, and all lie in the radius-$(h_0+1/2)$ ball about the image of r. Comparing Lebesgue volumes gives

$$\frac{N}{2} \le (2h_0+1)^d.$$

Hence

$$\begin{aligned} d &\ge \frac{\log(N/2)}{\log(2h_0+1)} \ge \frac{\log(N/2)}{\log(2H+3)} \\ &= (1-o(1))\frac{\log N}{\log(2+H)} = (1-o(1))q, \end{aligned}$$

where $H \to \infty$ was proved above. Thus $d \le q/2$ is impossible for all sufficiently large N, proving part (ii).

For Corollary 2, if $D \le (\log N)^\beta$ with $\beta < \kappa/2$, then $H \ge (\log N)^{1-\beta}$ and

$$q \asymp \frac{\log N}{\log\log N}.$$

It follows that $D = o(q^{\kappa/2})$, so Theorem 1 applies. □

Verification notes

The verification audit checked every occurrence in Section 4 of the source where N was taken sufficiently large in terms of fixed Δ. The degree enters only through:

1. the path-dependency count and inequality corresponding to (5);

2. the short-path comparison $m^{\kappa/2} \gg \log \Delta$;
3. the transition length $k_1 = \lfloor \log N / \log(\Delta - 1) \rfloor$ in (8) and the longer-path estimate.

All later estimates are degree-free, and (9) is exactly the uniform range they require. No finite computation is used as proof.

The main human-review focus should be the uniformity of the source's Lemma 3.3 in $k \leq k_0$ (the lemma is stated uniformly for all $k \geq 2$) and the interpretation of "complete tree" at non-full cardinalities. The lower bound is exact for full complete trees and uses the same truncation as the source for arbitrary N.

Limitations and novelty status

The theorem leaves open degrees for which $\log(\Delta - 1)$ is comparable to or larger than $q^{\kappa/2}$, does not optimize the constants, and does not address Problem 1.5. Its use of the thin-shell exponent means the displayed degree range improves automatically with any stronger uniform short-path estimate.

A bounded literature search on 9 August 2026 used arXiv:2504.15212, the exact paper title, the exact phrase "Trees of growing maximal degree", and keyword combinations involving geometric embeddings, trees, growing degree, and normed spaces. It found no separate paper answering Problem 1.6 or stating this uniformized range. Since the source v2 dates from 17 April 2026, novelty confidence is moderate rather than definitive.

Human-review recommendation

Review the three local-lemma comparisons and the Gaussian uniformity interval (9). If those checks pass, this packet gives a rigorous substantial partial answer to Problem 1.6 with an explicit threshold scale and universal constants on a nontrivial growing-degree regime.

References

[1] D. J. Altschuler, P. Dodos, K. Tikhomirov, and K. Tyros, *A universal threshold for geometric embeddings of trees*, arXiv:2504.15212v2 (17 April 2026), `https://arxiv.org/abs/2504.15212`.

[2] N. Alon and J. H. Spencer, *The Probabilistic Method*, 4th ed., Wiley, 2016.